

Department of CSE
United International University
CSE 1112: SPL
Evaluation 2 (Control Statements)

Time: 45 minutes

Marks: 10

ROLL:

MARK OBTAINED:

- | | | |
|----------|--|-----------|
| 1 | <p>After a busy day at UIU, you're making your way home. You unexpectedly fell over a dog's tail. Boom !! The entire pack of dogs begins to bark, and soon after, they begin chasing you! In order to avoid being attacked by dogs, you must run as fast as you can.</p> <p>Now, given the speed of n number of dogs, you are to find out the minimum speed that you must maintain so as not to get caught by any of them. Also, find the average speed of the dogs.</p> | 10 |
|----------|--|-----------|

Sample Input	Sample Output
Enter no. of dogs: 5 Enter Speed of dog no. 1: 20 Enter Speed of dog no. 2: 10 Enter Speed of dog no. 3: 12 Enter Speed of dog no. 4: 14 Enter Speed of dog no. 5: 21	Your speed should be a minimum of 21. Average speed of the dogs is: 15.4

Department of CSE
United International University
CSE 1112: SPL
Evaluation 2 (Control Statements)

Time: 45 minutes

Marks: 10

ROLL:

MARK OBTAINED:

1.	Shah Cement Factory has automated its stone crushing unit. A number of stones can be loaded into the unit at a time. However, noting down the max, min, and average weight of the stones to be loaded is important. However, before stones are loaded into the crushing unit, a sensor checks to see if the stone weights are within the limit. If the stone's weight is over 50 Kg it shouldn't be allowed into the crushing unit, "Stone beyond unit's max weight limit" should be printed and the program should be terminated without further calculation. Now, construct a program that takes input as shown and calculates the max, min, and average weight of the loaded stones. IT should also terminate the program when any value is above 50 as shown below. When no value > 50: <pre>Enter number of stones: 5 Enter 5 values (weights): 25.5 20.75 12.5 44.5 41.75 Max: 44.50, Min: 12.50, Avg:29.00 Total number of overweight stones: 2</pre> When any value > 50: <pre>Enter number of stones: 5 Enter 5 values (weights): 25 42 51.5 Stone beyond unit's max weight limit Process returned 0 (0x0) execution time : 8.970 s</pre>	10
----	---	----

Department of CSE
United International University
CSE 1112: SPL
Evaluation 2 (Control Statements)

Time: 45 minutes

Marks: 10

ROLL:

MARK OBTAINED:

2

Take a **4-digit integer** input “n” and print “YES” if the number is a **Digit-Prime**, otherwise print “NO”.

15

A number is a **Digit-Prime** if the sum of the digits of that number is a prime number. For example, if n is 2458 then the sum of the digits is (2+4+5+8 = 19) and 19 is a prime number. In this case, the output will be “YES”.

Again, If n is 3783 then the sum of digits is (3+7+8+3 = 21) and 21 is not a prime number. In this case, the output will be “NO”. Note that, you can not use arrays.

Sample Input	Sample Output
7345	YES
239	NO